Edward J. Carter

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Target Role

Senior Data Engineer

Summary

Senior data platform engineer with 10+ years of experience building scalable data systems that support analytics and machine learning workflows. Experienced in designing distributed data platforms, integrating batch and streaming pipelines, and enabling data access patterns for AI and advanced analytics.

Focuses on bridging data engineering and ML system requirements—supporting feature pipelines, hybrid retrieval patterns, and scalable data processing needed for modern AI-driven applications.

Core Skills

Data & AI Platform Engineering: Distributed data platforms, ML pipeline integration, feature and inference data systems

Processing & Compute: Apache Spark (PySpark), large-scale data processing, batch and streaming integration

Vector & Retrieval Systems: Qdrant, pgvector, embedding pipelines, hybrid retrieval (RAG patterns)

Streaming & Event Systems: Apache Kafka, real-time data pipelines, event-driven architectures

Data Quality & Governance: Validation frameworks, anomaly detection, schema enforcement

Platform Infrastructure: Kubernetes, containerized ML environments, Databricks

Data Storage: PostgreSQL, NoSQL systems, hybrid structured/unstructured data systems

Languages: Python, SQL

Professional Experience

Senior Data Engineer (Platform & Data Engineering) 12/2023 – Present
Ennoble First (SAFFIRE/CHINOOK)

- **Own and evolve enterprise data platform capabilities** supporting distributed processing, governed data access, and mission-critical analytics across engineering and operational teams
- **Define and implement data engineering standards** (ingestion, transformation, validation, lifecycle management), improving reliability and reducing rework across multiple teams and pipelines
- **Lead adoption of medallion-based architecture patterns** (Bronze/Silver/Gold), enabling consistent data lifecycle management and scalable, self-service data consumption
- **Establish data quality and validation frameworks**, improving trust, consistency, and usability of downstream datasets
- **Drive cross-functional alignment** across engineering, architecture, and operational stakeholders to translate mission requirements into scalable platform capabilities
- Improve accessibility and timeliness of enterprise data, enabling more reliable and responsive decision-making
- Contribute to technical strategy and proposal efforts, shaping solution architectures aligned with mission and organizational priorities
- Develop reusable architectural patterns that standardize distributed system design and support long-term platform evolution

Data Engineer 09/2023 – 12/2023
Los Alamos National Laboratory (Programming & Runtime Environment)

- **Designed and deployed secure, containerized compute environments** across classified and unclassified domains, enabling scalable data processing and experimentation workflows
- **Architected microservice-based systems** supporting HPC resource utilization and license management, improving system reliability and operational efficiency
- **Led Kubernetes-based platform deployment and standardization**, increasing team productivity and enabling consistent, reproducible infrastructure delivery
- Contributed to platform engineering practices improving system observability, scalability, and operational consistency

Software Engineer / Data Engineer 05/2021 – 12/2023
Army Software Factory (Army Futures Command)

- **Led development and optimization of large-scale data pipelines processing 500M+ records**, reducing runtime from 24 hours to ∼26 minutes and enabling near real-time decision support
- Designed and delivered curated datasets into enterprise analytics platforms (Army Vantage), improving operational visibility and data accessibility across engineering and leadership stakeholders
- **Drove modernization of fragmented data systems into unified, scalable architectures**, increasing interoperability and enabling enterprise-wide data access
- Partnered across engineering, operations, and product stakeholders to align data platform capabilities with mission objectives and long-term priorities

Data Engineer 04/2021 – 09/2022
Texas ProTax

- Designed and implemented data integration workflows across SQL and NoSQL systems, enabling scalable pipelines for analytics and machine learning initiatives
- Delivered curated, analysis-ready datasets aligned with governance and modeling standards, improving data usability and accelerating downstream decision-making

Digital Intelligence Systems Master Gunner (DISMG) 03/2017 – 04/2021
United States Army

- **Led integration and modernization of enterprise intelligence data platforms** (DCGS-A CD2, Palantir-powered), aligning data architecture with mission-critical decision systems
- Directed cross-functional teams spanning analysts, engineers, and operational personnel to support data-driven mission execution
- **Owned reliability, accuracy, and availability of mission-critical data systems**, enabling consistent real-time decision support in operational environments
- Drove adoption of data governance standards and structured data practices across distributed teams

Intelligence Analyst / Data Analyst → Data Scientist → Data Architect 09/2002 – 04/2024
United States Army

- Progressed through increasingly senior roles, developing end-to-end ownership of the data lifecycle from ingestion and modeling to advanced analytics and decision support
- Designed scalable data architectures and standardized analytical workflows, enabling consistent and repeatable decision-making across distributed environments
- Served as a technical bridge between analysts, engineers, and leadership stakeholders, translating operational needs into scalable data solutions
- **Led and mentored personnel across multiple organizational levels**, contributing to workforce development, technical capability growth, and mission effectiveness

Representative Platform Systems

Large-Scale Medallion Data Platform *PySpark, Delta Lake, Parquet*

- *Architecture:* Medallion (Bronze/Silver/Gold) data platform with integrated data quality and incremental processing
- Designed and implemented an end-to-end data platform standardizing ingestion, transformation, and consumption across analytical workloads
- Established data quality validation and investigation patterns, improving trust and usability of downstream datasets
- Optimized transformation and partitioning strategies to support scalable analytics and machine learning workloads
- **Impact:** Reduced friction in accessing analysis-ready data and improved consistency of downstream datasets, enabling more reliable and timely decision-making

Hybrid Streaming & Batch Data Platform *Kafka, Spark, Python, Kubernetes*

- *Architecture:* Hybrid platform combining event-driven streaming with batch processing layers
- Designed ingestion and processing patterns supporting scalable, modular data transformation across real-time and batch workloads
- Implemented incremental processing and validation strategies aligned to modern data platform design
- Enabled consistent data flow patterns supporting downstream analytics and operational systems
- **Impact:** Reduced data latency and improved consistency between real-time and batch systems, enabling more timely and reliable downstream consumption

Geospatial Data Platform (DoD Context) *Databricks, Apache Sedona, H3, AWS*

- *Architecture:* Distributed geospatial data platform leveraging medallion modeling and spatial indexing (H3)
- Designed scalable geospatial processing workflows supporting large-scale spatial joins and query workloads

- Integrated spatial indexing and geospatial data models to enable high-performance spatial analytics
- Enabled map-based analytical workflows supporting operational decision-making at scale
- **Impact:** Improved performance and accessibility of geospatial analysis, enabling faster exploration and more responsive operational insights

Education & Certifications

- M.S. Data Engineering, Auburn University (Expected 2027)
- B.S. Database Management / Data Analytics, Western Governors University
- ITILv4 — CIW-Web Foundations — CIW-Data Analyst — CompTIA Project+